

KOLYMSKOYE, Russian Federation

WMO: 251210

Lat: 68.729N

Lon: 158.704E

Elev: 39

StdP: 14.68

Time Zone: 11.00 (E11)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.4	-46.7	-53.9	0.2	-48.9	-51.2	0.3	-46.4	18.9	-7.5	17.5	-7.3	3.1	180	0.915

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	75.6	60.6	71.1	58.1	67.2	56.4	62.1	71.2	59.8	68.3	57.5	65.8	7.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.5	73.3	65.8	55.8	66.4	63.2	52.9	59.7	61.9	28.0	71.5	26.3	68.3	24.8	65.9	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.7	15.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-51.8	66.0	3.7	2.7	-54.4	67.9	-56.6	69.4	-58.6	70.9	-61.3	72.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-26.4	-24.5	-14.5	8.1	26.8	46.6	52.4	48.7	38.0	16.9	-8.1	-22.3		
	DBStd	31.26	12.25	12.56	13.14	11.63	10.90	9.89	8.35	7.78	6.90	11.13	14.00	14.36		
	HDD50	14182	2369	2086	1998	1257	720	183	67	120	370	1027	1743	2242		
	HDD65	19362	2834	2506	2463	1707	1184	554	404	507	811	1492	2193	2707		
	CDD50	313	0	0	0	0	1	81	141	81	9	0	0	0		
	CDD65	18	0	0	0	0	0	3	12	3	0	0	0	0		
	CDH74	195	0	0	0	0	0	51	123	21	0	0	0	0		
	CDH80	30	0	0	0	0	0	6	23	1	0	0	0	0		
	WSAvg	7.5	6.2	6.5	6.8	7.2	8.3	8.7	7.9	8.1	8.0	7.1	7.1	7.6		
	PrecAvg	13.7	0.7	0.6	0.6	0.4	0.6	1.2	1.9	1.8	1.7	1.7	1.4	0.9		
(15) Precipitation	PrecMax	21.2	1.7	2.2	1.4	1.0	1.1	2.3	3.4	3.4	4.1	3.9	4.9	2.2		
	PrecMin	8.8	0.1	0.1	0.1	0.0	0.1	0.4	0.3	0.5	0.2	0.6	0.4	0.2		
	PrecStd	3.3	0.4	0.4	0.4	0.2	0.2	0.5	0.9	0.7	0.9	0.8	0.8	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	15.7	20.0	38.6	59.9	79.8	82.9	76.6	64.4	39.9	28.4	21.5		
		MCWB	12.1	14.1	17.9	33.2	48.9	58.7	64.2	63.0	52.8	38.2	27.7	20.5		
	2%	DB	2.4	4.0	14.8	35.8	52.0	74.2	76.8	71.4	57.8	35.4	23.9	9.9		
		MCWB	1.8	3.5	13.2	31.2	43.3	58.5	60.7	59.4	50.5	34.5	23.0	9.5		
	5%	DB	-3.4	-2.4	10.4	31.8	46.9	70.0	72.3	67.4	52.6	33.2	17.7	2.4		
		MCWB	-3.9	-2.8	9.0	28.7	39.8	55.4	59.2	57.5	46.1	32.7	17.2	1.9		
	10%	DB	-9.2	-6.4	5.2	26.5	42.7	65.4	67.4	62.7	48.2	31.6	11.7	-2.2		
		MCWB	-9.7	-6.8	4.4	23.7	37.4	53.2	57.4	55.7	43.6	31.1	11.3	-2.7		
	0.4%	WB	12.5	14.3	18.3	34.1	49.3	62.6	66.0	64.7	54.3	38.5	27.8	20.7		
		MCDB	13.1	15.9	19.9	37.7	59.5	74.3	76.7	73.1	62.4	40.1	28.3	21.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	1.9	3.5	13.4	31.5	43.8	58.6	62.6	62.0	51.0	34.8	23.2	9.3		
		MCDB	2.4	4.0	14.9	35.3	51.4	71.6	72.9	69.4	56.6	35.3	23.9	9.7		
	5%	WB	-3.7	-2.9	9.2	28.3	40.3	56.7	60.4	59.3	47.3	32.9	17.2	1.9		
		MCDB	-3.2	-2.4	10.5	32.0	46.3	68.7	68.8	65.9	51.9	33.3	17.7	2.2		
	10%	WB	-9.6	-6.8	4.4	23.7	37.3	54.1	58.3	56.0	43.6	31.2	11.4	-2.7		
		MCDB	-9.2	-6.5	5.2	26.3	42.4	64.5	66.5	61.6	47.3	31.5	11.8	-2.3		
	5% DB	MDBR	11.0	10.8	15.9	18.5	13.9	16.4	14.3	14.3	9.8	8.4	11.9	11.5		
		MCDBR	16.8	16.9	20.1	20.7	19.6	25.2	22.9	24.1	18.0	7.5	14.8	18.6		
		MCWBR	16.5	16.6	18.6	17.3	12.0	12.4	9.9	12.3	10.7	6.9	14.6	17.7		
		MCWBR	17.0	16.4	19.5	20.3	19.2	24.4	21.2	20.4	16.5	7.1	15.0	17.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	16.7	16.2	18.1	17.1	12.2	13.4	11.4	12.7	10.8	6.6	14.8	17.3		
(40) Clear-Sky Solar Irradiance	taub		0.166	0.221	0.288	0.336	0.374	0.357	0.415	0.389	0.317	0.255	0.182	N/A		
	taud		1.920	2.063	2.072	2.019	2.186	2.430	2.255	2.296	2.461	2.144	1.922	N/A		
	Ebn at Noon		27	180	228	250	255	267	242	234	228	157	16	0		
	Edh at Noon		2	14	27	39	37	30	35	30	19	13	2	0		
(44) All-Sky Solar Radiation	RadAvg		7	148	692	1453	1854	1819	1543	1077	557	180	25	0		
	RadStd		1	16	31	67	146	142	155	107	62	31	3	0		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+3.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page